

Step 1: Create Extract Job

1. Go to the AWS Glue service
2. On the left pane click on **"ETL jobs"**
3. Click **"Script editor"**
4. Engine **"Spark"**
5. Options **"Upload script"**
6. Then upload the script called **"extract_job.py"**
7. Click on Job details
8. Name: **order-extract-job-[your-name]**
9. IAM: Confirm your IAM role
10. Worker type: G 1X
11. Requested number of workers: 2
12. Scroll down click **"Advanced properties"**
13. Scroll down to **Job parameters**
14. Click **Add new parameter** and add the below:
 - Key: **--input_path**
 - Value: **s3://BUCKET-NAME/datafiles/raw/sales_data.csv**
 - Key: **--output_path**
 - Value: **s3://BUCKET-NAME/datafiles/processed/extracted/**
15. Then click "Save" DO NOT RUN.

Step 2: Create Validation Job

1. On the left pane click on **"ETL jobs"**
2. Click **"Script editor"**
3. Engine **"Spark"**
4. Options **"Upload script"**
5. Then upload the script called **"validation_job.py"**
6. Click on Job details
7. Name: **order-validate-job-[your-name]**
8. IAM: Confirm your IAM role
9. Worker type: G 1X
10. Requested number of workers: 2
11. Scroll down click **"Advanced properties"**
12. Scroll down to **Job parameters**
13. Click **Add new parameter** and add the below:
 - Key: **--input_path**
 - Value: **s3://BUCKET-NAME/datafiles/processed/extracted/**
14. Then click "Save" DO NOT RUN.

Step 3: Create Aggregate Job

1. On the left pane click on **"ETL jobs"**
2. Click **"Script editor"**
3. Engine **"Spark"**

4. Options **"Upload script"**
5. Then upload the script called **"aggregate_job.py"**
6. Click on Job details
7. Name: **order-aggregate-job-[your-name]**
8. IAM: Confirm your IAM role
9. Worker type: G 1X
10. Requested number of workers: 2
11. Scroll down click **"Advanced properties"**
12. Scroll down to **Job parameters**
13. Click **Add new parameter** and add the below:
 - Key: --input_path
 - Value: **s3://BUCKET-NAME/datafiles/processed/extracted/**
 - Key: --output_path
 - Value: **s3://BUCKET-NAME/datafiles/processed/aggregated/**
14. Then click "Save" DO NOT RUN.

Step 4: Create Workflow

1. On the left pane click on **"Workflows (orchestration)"**
2. Click **"Add workflow"**
3. Workflow name: **sales-etl-workflow-[your-name]**
4. Once successfully created click on your workflow

Step 5: Add Trigger to Workflow

1. On the Graph UI click **"Add trigger"**
2. Click **"Add new"**
3. Name: **start-extract-trigger-[your-name]**
4. Trigger type: On demand
5. Click **"Add"**

Step 6: Add Extract Job Node

1. On the Graph UI click "Add node"
2. Select the **order-extract-job-[your-name]**
3. Click "Add"

Step 7: Add Trigger After Extract Job

1. Select the **order-extract-job-[your-name]** then click "Add trigger"
2. Name: **validate-after-extract-[your-name]**
3. Trigger type: Event
4. Trigger logic: Start after ANY watched event
5. Click "Add"

Step 8: Add Validation Job Node

1. Click on **"Add node"** next to the trigger you just added
2. Select the `order-validate-job-[your-name]` you created earlier
3. Click "Add"

Step 9: Add Trigger After Validation Job

1. Select the `order-validate-job-[your-name]` then click "Add trigger"
2. Name: `aggregate-after-validate-[your-name]`
3. Trigger type: Event
4. Trigger logic: Start after ANY watched event
5. Click "Add"

Step 10: Add Aggregate Job Node

1. Click on **"Add node"** next to the trigger you just added
2. Select the `order-aggregate-job-[your-name]` you created earlier
3. Click "Add"

Step 11: Review and Run Workflow

1. Review your workflow
2. Scroll up then click **"Run workflow"**

Step 12: Monitor Workflow Execution

1. Next to the Graph tab, click **"History"**
2. On the Workflow runs click the radio button next to Workflow run ID
3. Then click **"View run details"**
4. Scroll down and review your running job on the graph.
5. Once all nodes are **GREEN**, proceed to next step

Step 13: Download and Review Results

1. Go to your S3 bucket: `s3://BUCKET-NAME/datafiles/processed/aggregated/`
2. Download and review the .csv